Design and Fabrication of Vapour Compression Based Mobile Water Chilling Unit

A report

Presented

To

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Declaration of Academic Honesty and Integrity

We, Sritam Mohanty, Arya Raj and Yuvraj Singh, hereby declare that this project report titled "Design and Fabrication of Vapour Compression Based Mobile Water Chilling Unit" submitted to the Department of Mechanical Engineering, National Institute of Technology Jamshedpur, in fulfillment of the requirements for the Bachelor of Technology degree, is our original work. We have not plagiarized any part of this report from other sources, and all references have been duly acknowledged. This work has not been submitted elsewhere for the award of any other degree.

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This certifies that the thesis entitled "**Design and Fabrication of Vapour Compression based Mobile Water Chilling Unit**" submitted to the National Institute of Technology Jamshedpur by Mr. Sritam Mohanty, Arya Raj and Yuvraj Singh for the award of B.Tech. Degree in Mechanical Engineering does not contain any classified information. This work is original and has not been submitted to any institution or university for the award of any degree.

Signature of the candidates
Date:

(.....)
(HOD/ Faculty Advisor)
Date:

(.....)
Dr. Shashank Pandey
Date:

PREFACE

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This project report documents our efforts to design and develop a vapour compression based mobile water chilling unit as part of the Major Project course (ME1801) during the Spring Semester of 2026. The primary motivation behind this project was to address the reusability of components to design a water chilling unit with mobility. Through this work, we aimed to create a practical, efficient, and portable solution that ensures chilled water delivery using eco-friendly refrigerants while using low-cost reusable components. This report outlines our design process, methodology, results, and conclusions, providing a comprehensive overview of our investigation.

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to **Dr Shashank Pandey** for his invaluable guidance, unwavering support, and encouragement throughout the course of this minor project. His insightful suggestions and expertise have been instrumental in shaping the project and ensuring its successful completion.

We are deeply thankful for his patience and for providing us with the opportunity to explore and develop our skills in this domain. His mentorship has not only enhanced our technical knowledge but also inspired us to strive for excellence.

Once again, We extend our sincere thanks to Dr Shashank Pandey for his constant motivation and for being a source of inspiration throughout this journey.

Date:

Place: Jamshedpur

ABSTRACT

Availability of cool drinking water during intense summer conditions is a challenge and a necessity of relief. The project focuses on developing, designing and fabricating a mobile vapour compression-based water chilling unit using readily available and economical components. The system comprises of domestic refrigerator compressor, a water dispensing water tank which acts as the cooling chamber, embedded within it are the copper tubes which act as the pathway for the refrigerant which cool the water and a condenser to act as heat exchanger. The refrigerant used is R-600a (Isobutane) which has minimal harsh effects on the surrounding environment considering its good thermodynamic properties, zero ozone depletion potential and low global warming potential, making it a good alternative to other refrigerants.

The primary objective of this project is to develop an experimental setup and as a model to combat existing limitations of larger water chilling units, high energy consumption and wastage, limited mobility while using lesser environmentally friendly refrigerants. This setup elaborates on the efficiency of vapour compression refrigerating system which acts as an ideal system to cool a large quantity of water in the best efficient manner possible.

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LIST OF ABBREVIATIONS

VCRS: Capour Compression Refrigeration System

ODP: Ozone Depletion Potential

GWP: Global Warming Potential

CHAPTER 1

INTRODUCTION

1.1 Background

Water is an indispensable factor in our day-to-day lives. With the percentage of drinkable water even being a smaller part of the overall water content present. The need for a cool drinking water solution to the larger masses in the intense summer heat presents a challenge which needs to combine factors like a progressive and steady rate of providing cool water, the ability to maintain the water temperature over a long period of time and be efficient and a cost affordable apparatus at the same time.

To combat this, Cooling systems have been integrated in our everyday lives. With applications ranging from household activities like food preservation, air conditioning to industrial cooling and thermal support systems. Various means of cooling have been developed depending on the scale of cooling required.

The purpose of this project to look into the cooling systems required to provide chilled water to the larger masses with the additional advantage of mobility to allow the apparatus to be moved to the appropriate location to provide chill water as and when required. Additional focus has been put into using zero-ODP refrigerant to be compliant as an environment friendly apparatus to provide cool water.

The project has its main applications in the intense summer conditions to provide chilled water to the masses while also acting as an experimental setup to look into the efficiency of cooling provided using the concepts learned in mechanical engineering domain. It looks into the aspect of using low cost, reusable components to provide a water chilling apparatus which can be used in households and small end businesses.

1.2 Problem Statement

However, there are various problems associated with the need to provide chilled water during the intense summer heat conditions:

- Water cooling and dispensing units are usually stationary, placed at a singular space even if the option of mobility is provided.
- The traditional water cooling and dispensing units are quite large in size which does raise the concern of portability and mobility,
- Usage of Refrigerant with high GWP which raises concern regarding environmental conservation.
- Frequent Compressor Cycling which leads to high energy consumption and energy wastage in the longer term.

These challenges demand the necessity for a portable, mobile and cost-efficient water-cooling unit which uses a good chilling unit with an environment friendly refrigerant to reduce the impact on environment.

1.3 Objectives

The objectives of this project are:

1. To create a water chilling unit that is efficient in cooling water in the minimum time possible considering energy consumption and prevent energy wastage.
2. Use a refrigerant with minimal ODP and GWP to reduce the impact of the water chilling unit on the surrounding environment.
3. To design a cart which acts as the source of mobility for the entire chilling unit which allows it to be moved to a desired location.

1.4 Scope of Work

The project encompasses the following aspects:

- A literature survey of the existing cooling techniques in modern world with focus on the components involved in the chilling process.

- Inclusion of a VCRS unit which acts as the most efficient solution to chill water in the minimal time.
- Using R-600a (Isobutane) as the refrigerant which has minimum ODP and GWP when compared to other refrigerants as per the literature survey.

CHAPTER 2
LITERATURE SURVEY

2.1 Literature Review

The improvement of performance of Vapour Compression Refrigeration Systems has been a scope of major research area due to increasing demand inefficient cooling systems and the global environmental concerns associated with conventional refrigerants. In the recent years, hydrocarbon refrigerants like R-600a have gained major traction because of their properties like minimal Global Warming Potential, zero Ozone Depletion Potential and favourable thermodynamic properties. This project which inculcates a refrigerator compressor, copper tubing network with dyer, condenser and R-600a refrigerant for applications in water chilling, aligns with current sustainable refrigeration standards.

Many researchers have investigated the suitability and applications of R-600a as an alternative refrigerant in domestic refrigeration systems. Recent works in experimental domain have reported that R-600a provided a performance comparable to R-134a with improvements in terms of energy efficiency and lower impact o environment. Thus, making it a sustainable replacement for applications in household refrigeration.

Kazmi et al. (2025) experimentally studied R-600a/R-290 blends in refrigeration-freezer systems based in households and concluded that optimized hydrocarbon blends could maintain the performance of the system even at lower compressor speeds, thereby resulting in a reduction in energy consumption. This shoes the effectiveness of hydrocarbon refrigerants in improving the efficiency of refrigeration systems.

Thermodynamic studies which compared R-600 and R-600a with conventionally used refrigerants, showed that the hydrocarbon refrigerants exhibited superior results in terms of energetic and exergic performance. Thus, researchers concluded that the minimization of temperature difference between condenser and evaporator improves overall COP, while R-600a showed superior efficiency in domestic based refrigeration systems.

Design of condenser is another important factor which affects VCRS performance. Research and studies on alternative geometries of condensers which uses serpentine and spiral tube condensers revealed the fact that the surface area of condenser and the heat transfer characteristics involved, highly influence the cooling performance and the efficiency of the system.

Mahesh et al. (2020) investigated enhancement in refrigeration performance through improvements in condenser cooling and recovery of waste heat. The results indicated that insufficiency in condenser cooling reduces the COP of the system, emphasizing the need and importance of improving the effectiveness of heat rejection devices like finned radiators and compact heat exchangers.

Additionally, experimental studies on VCRS units which have been modified using R-600a as refrigerant highlight the importance of moisture-free flow of refrigerant. The installation of filter dryers in copper tubing systems prevents the accumulation of moisture, blockages in capillary tubes and internal contamination. Thereby, resulting in overall improvement in system reliability and lifespan.

From the reviewed literature, it is to be noted that although extensive studies in terms of research and experimentation, exist on R-600a refrigeration systems, limited work has been cited on combining a refrigerator compressor, condenser and a water dispensing evaporator unit into a compact, mobile water chilling prototype. Thus, the project aims to bridge this cap by developing a low-cost, eco-friendly and efficient VCRS model suitable for education purposes through experimentation and small-scale cooling applications.

2.2 Components

2.2.1 Compressor

The compressor is the heart of the system. It extracts the low-pressure, low temperature refrigerant vapour from the evaporator and compresses it into high pressure, high temperature vapour.



Fig 2.1: Compressor

2.2.2 Copper Tubing

Copper tubes act as the pathway for refrigerant flow throughout the system. Copper is preferred due to its properties like high thermal conductivity, corrosion resistance, and ease of malleability and brazing.



Fig 2.2: Copper Tubing

2.2.3 Filter Dryer

A dryer is included within the copper tubing line, generally after the position of condenser. Its purpose is the removal of moisture, contaminants and dirt from the refrigerant vapour. Moisture can possibly freeze in capillary tube and reduce effective performance, so the filter dryer is an essential component for reliable and sustained operation.



Fig 2.3: Filter Dryer

2.2.4 Condenser

The high-pressure hot refrigerant vapour enters the condenser, where heat is rejected to the immediate surroundings. As the heat is removed from the refrigerant vapour, the refrigerant condenses into a high-pressure liquid.



Fig 2.4: Spiral Tube Condenser

2.2.5 Expansion Device

The liquid refrigerant is passed through a capillary tube or expansion valve where the pressure drops suddenly, causing partial evaporation and a sharp decrease in temperature.



Fig 2.5: Capillary tube

2.2.6 Evaporator

The cold low-pressure refrigerant absorbs heat from the water and the surrounding chamber and evaporate into vapour.



Fig 2.6: Spiral Coil Evaporator

CHAPTER 3

METHODOLOGY

3.1 Design Process

3.1.1 Initial Design

The main challenge was to find a suitable chilling unit solution which can chill water in the minimal time, be efficient in doing so while maintaining the concern for energy consumption and energy wastage.

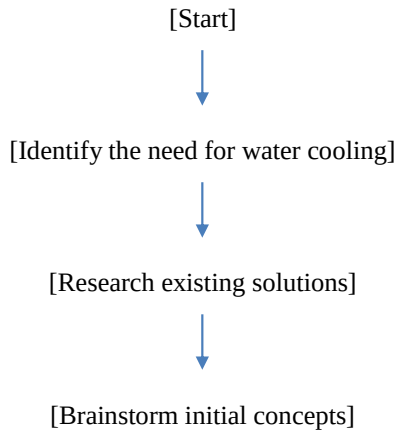
Converging on two chilling techniques i.e., Peltier Cooling and VCRS Cooling. Although low costing, Peltier cooling took longer times to cool large quantities of water with inconsistencies in the temperature drop and cooling efficiency.

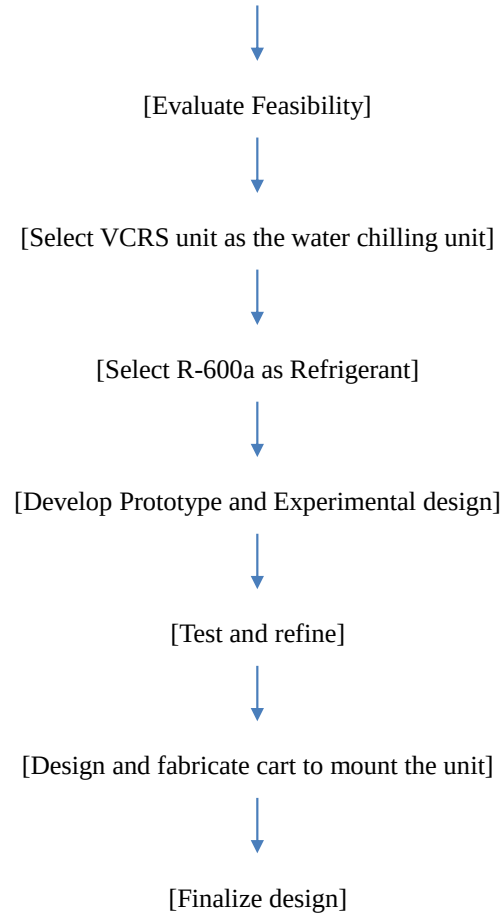
In initial design, Peltier cooling was considered as the chilling unit but due to complexity in temperature control and above cited reasons was found impractical during team discussions.

3.1.2 Final Design

The final design incorporates a VCRS cooling unit with R-600a (Isobutane) as the refrigerant with the minimum ODP and GWP mounted on the top of a cart which provides the necessary mobility. Further Inclusion of an inverter and battery to make it self-sufficient in cases of unavailability of power source in any location.

Flow Chart of Design Process:





3.2 Working Principle

3.2.1 VCRS Cycle

The water chilling unit consists of a compressor, a condenser, copper tubing with a filter dryer and R-600a as refrigerant. The entire chilling unit acts on a VCRS cycle. A VCRS cycle is widely used is current day refrigeration units, air conditioner units and thermal cooling-based units in household and industrial systems. The system operates on the principle of removing heat from a low temperature region and rejecting it a higher temperature region by circulating a refrigerant to act as the working fluid.

3.2.2 Working Process of the Machine

The compressor acts as the heart of the system. It draws the low pressure, low heat refrigerant from the evaporator, compresses it into high temperature, high pressure vapour and cycles it through the system. Copper tubes act as the path for the refrigerant, preferred for its good thermal conductivity and its malleable and ductile properties to be formed into tubes and bended as per requirements.

The high pressure, high temperature refrigerant vapour goes into the condenser which rejects heat form the refrigerant converting it into a high-pressure liquid. It then goes into a capillary tube or an expansion valve where there is a sudden decrease in pressure, causing partial evaporation and temperature drop. A dryer is installed within the copper tubing which prevents any moisture, dirt and impurities to get into the refrigerant which can impede the cooling efficiency of the apparatus.

The refrigerant cycles through the apparatus cooling the water in the water dispensing tank through the copper tubes. There is a cutoff on the compressor when there is excessive load on the compressor it will stop the supply of refrigerant through the apparatus and resume once again when needed and when the load reduces.

3.3 Guide to make a Vapour Compression Based Mobile Water Chilling Unit

1**. Gather Materials and Tools**

Materials:

- R-600a rated Compressor (Power rated – 110 watts)
- Steel water dispensing unit
- Copper tubing for evaporator
- Finer Copper tubing for capillary tubes
- Metal Base Plate (2.25 ft * 1 ft)
- Spiral Condenser
- Steel square pipes
- Wheels (7.5 cm diameter)
- Electric Wiring

- Screws, Bolts and Nuts
- Power Source (AC adapter)
- R-600a refrigerant

Tools:

- Welding Machine
- Angle Grinder
- Drill Machine
- Measuring Tape
- Cutting Tools

2. Build the Vapour compression-based water chilling unit

Setup the metal base plate on a stable surface. Place the water dispensing unit on one end and weld it to that position.

Place the compressor in the center and weld it into position. Then the copper tubes are beaten into shape of coil for evaporator and the tubes are connected first from inlet and outlet to the water dispensing unit where the evaporator is placed, till the condenser, completing a full cycle of flow of refrigerant.

Then a filter dryer is connected to the copper tube and a finer copper capillary tube is placed after the condenser which acts as the expansion tube.
After this, the R-600a refrigerant is put into the compressor and the wiring is done for the compressor to power the apparatus through an AC adapter.

3.Build the Frame

The steel square pipes are cut into the required dimension (760 mm * 610 mm * 460 mm). The base rectangular frame is welded first. Then the height is joined by welding the vertical pipes.

Make the top frame horizontal to give its metallic part of box structure.

Attach the wheels to the frame.

4.Place the water chilling unit on the frame

Place the base plate of the water chilling unit on the top of the frame.

Then the battery and inverter placed on the lower part of the frame to balance the weight aspect of the apparatus.

5.Wiring and Controls

All the wirings should be properly insulated.

Retain the AC adapter and connect the apparatus to proper rated connection.

6.Test the apparatus

A certain quantity of water was put into the water dispensing unit and the machine is turned on.

Then it is seen the water is cooled down after some time.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Performance Evaluation

4.1.1 Evaluation Process

The temperature was measured at two points in the apparatus which are the evaporator temperature and the condenser temperature using a thermocouple. The properties of enthalpy at each point of VCRS cycle was taken from the refrigeration table of R-600a refrigerant table.

Assumptions:

1. The rated compressor power was taken as rated input of power.
2. The vapour is considered to saturated at evaporator and saturated liquid at condenser.
3. There is no superheating and subcooling at any point.
4. Ideal (isentropic) compression is considered for calculation processes.

4.1.2 Calculation Process

Taking data from (table mentioned later), the calculation steps are mentioned below:

Step 1: Identify the cycle states

STATE	DESCRIPTION
STATE 1	Compressor inlet (saturated vapour at T_e)
STATE 2	Compressor outlet
STATE 3	Condenser inlet (saturated liquid at T_c)
STATE 4	After expansion valve

Step 2: Get the properties from R-600a table

At T_e = 20°C (Evaporator)

From saturation table:

- $P_1 \approx 2.9$ bar
- $h_1 \approx 408$ kJ/kg
- $s_1 \approx$ constant (used later)

At Tc = 35°C (Condenser)

From saturation table:

- $P_2 \approx 5.4$ bar
- $h_3 \approx 255$ kJ/kg

Step 3: Expansion process (3 → 4)

This is a throttling process, so: $h_4 = h_3$
Therefore:

- $h_4 = 255$ kJ/kg

Step 4: Compression process (1 → 2)

Assuming isentropic compression, so: $s_2 = s_1$

Now:

- Go to superheated table at $P_2 = 5.4$ bar
- Find enthalpy where entropy = s_1

From tables:

- $h_2 \approx 432$ kJ/kg

Step 5: Compressor work

$W_c = \dot{m}(h_2 - h_1)$

We rearrange to find mass flow rate: $\dot{m} = \frac{W_c}{h_2 - h_1}$

Given:

W_c = 110 W = 0.11 kW

$h_2 - h_1$ = 432 − 408 = 24 kJ/kg

$$\dot{m} = \frac{0.11}{24} = 0.0046 \text{ kg/s}$$

Step 6: Cooling capacity

$$Q_e = \dot{m}(h_1 - h_4)$$

$h_1 - h_4$ = 408 − 255 = 153 kJ/kg

$$Q_e = 0.0046 \times 153 = 0.70 \text{ kW}$$

Step 7: COP calculation

$$COP = \frac{Q_e}{W_c}$$

$$COP = \frac{0.70}{0.11} \approx 6.4$$

The above steps are followed to fill the data table and accordingly plot the data points to measure the performance of the apparatus considering the assumptions taken for calculative processes.

4.2 Datapoints and Plots

4.2.1 Data Table

Te (°C)	Tc (°C)	P1 (bar)	P2 (bar)	h1	h2	h3	h4	mass flow rate (kg/s)	Qe (kW)	Wc (kW)	COP
23	32	3.2	4.8	410	430	250	250	0.0055	0.88	0.11	8
22	33	3.1	5	409	431	252	252	0.005	0.78	0.11	7.1
21	34	3	5.2	408	432	253	253	0.0046	0.71	0.11	6.5
20	35	2.9	5.4	408	432	255	255	0.0046	0.7	0.11	6.4
19	36	2.8	5.6	407	433	257	257	0.0042	0.63	0.11	5.7
18	36	2.7	5.6	406	433	257	257	0.0041	0.61	0.11	5.6
17	37	2.6	5.8	406	434	258	258	0.0039	0.58	0.11	5.3
16	37	2.5	5.8	405	434	258	258	0.0038	0.56	0.11	5.1
16	38	2.5	6	405	435	260	260	0.0037	0.54	0.11	4.9
15	39	2.4	6.2	405	436	262	262	0.0035	0.5	0.11	4.5
15	40	2.4	6.4	405	437	263	263	0.0034	0.48	0.11	4.4
15	41	2.4	6.6	405	438	264	264	0.0033	0.47	0.11	4.3
15	42	2.4	6.8	405	439	265	265	0.0032	0.45	0.11	4.1
15	43	2.4	7	405	440	266	266	0.0031	0.43	0.11	3.9
14	44	2.3	7.2	404	441	267	267	0.003	0.41	0.11	3.7
13	45	2.2	7.4	403	442	268	268	0.0029	0.39	0.11	3.5
12	45	2.1	7.4	402	442	268	268	0.0028	0.38	0.11	3.4
11	45	2	7.4	401	442	268	268	0.0027	0.36	0.11	3.3
10	45	1.9	7.4	400	442	268	268	0.0026	0.34	0.11	3.1
9	45	1.8	7.4	399	442	268	268	0.0025	0.33	0.11	3
8	45	1.7	7.4	398	442	268	268	0.0024	0.31	0.11	2.9
7	45	1.6	7.4	397	442	268	268	0.0023	0.3	0.11	2.8
6	45	1.5	7.4	396	442	268	268	0.0022	0.28	0.11	2.6
5	45	1.4	7.4	398	440	265	265	0.0026	0.35	0.11	3.1
4	45	1.3	7.4	397	441	266	266	0.0024	0.31	0.11	2.8
3	45	1.2	7.4	396	441	266	266	0.0023	0.3	0.11	2.7
2	45	1.1	7.4	395	441	266	266	0.0022	0.28	0.11	2.6
1	45	1	7.4	395	442	267	267	0.0021	0.27	0.11	2.5
0	45	0.9	7.4	395	442	267	267	0.0023	0.3	0.11	2.8
-1	45	0.85	7.4	394	443	268	268	0.0022	0.28	0.11	2.6
-2	45	0.8	7.4	393	444	269	269	0.0021	0.26	0.11	2.4
-3	45	0.75	7.4	392	445	270	270	0.002	0.24	0.11	2.2
-4	45	0.7	7.4	391	446	271	271	0.0019	0.23	0.11	2.1
-5	45	0.65	7.4	390	447	272	272	0.0018	0.21	0.11	1.9
-5	46	0.65	7.6	390	448	273	273	0.0018	0.21	0.11	1.9
-6	46	0.6	7.6	389	450	274	274	0.0017	0.2	0.11	1.8
-5	47	0.65	7.8	390	449	274	274	0.0018	0.21	0.11	1.9
-5	48	0.65	8	390	450	275	275	0.0017	0.2	0.11	1.8
-5	49	0.65	8.2	390	451	276	276	0.0017	0.19	0.11	1.7
-5	50	0.65	8.4	390	452	277	277	0.0016	0.18	0.11	1.6
-5	51	0.65	8.6	390	453	278	278	0.0016	0.18	0.11	1.6
-6	52	0.6	8.8	389	455	279	279	0.0015	0.17	0.11	1.5
-5	53	0.65	9	390	454	280	280	0.0016	0.18	0.11	1.6
-5	54	0.65	9.2	390	455	281	281	0.0015	0.17	0.11	1.5
-5	55	0.65	9.4	390	456	282	282	0.0015	0.16	0.11	1.45
-6	56	0.6	9.6	389	458	283	283	0.0014	0.16	0.11	1.4

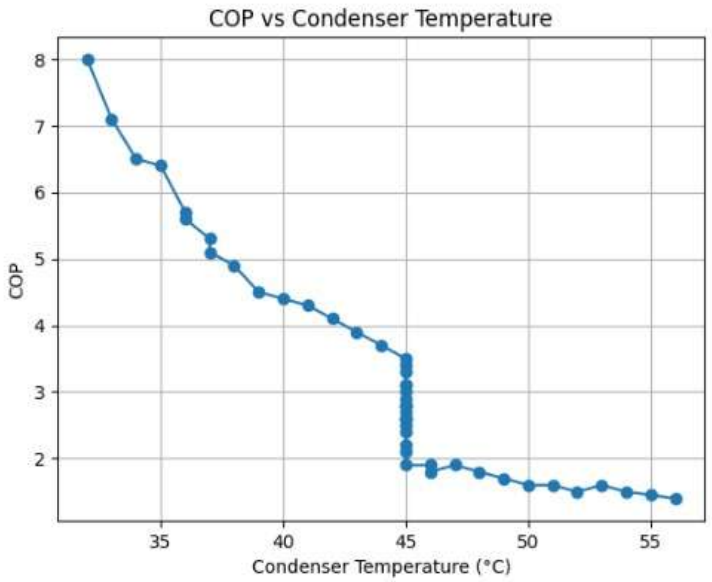
4.2.2 Data Plots

The motive of collecting data points is to plot a graph and derive a relationship between the various data categories. The main plots we considered are:

1. COP vs Condenser Temperature
2. COP vs Evaporator Temperature
3. COP vs Temperature Difference
4. Q_e vs Evaporator Temperature
5. COP vs Pressure Ratio

These plots are the major plots which describe the major relationship between the data points and additionally detail on the performance and effectiveness of the apparatus as a water chilling unit at different temperatures.

1. **COP vs Condenser Temperature**



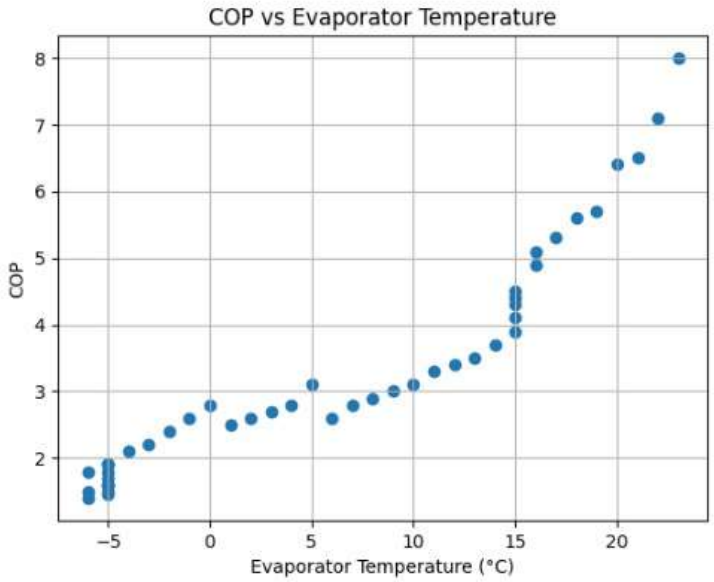
The graph between COP and Condenser Temperature demonstrates an inverse relationship between these two factors. As the condenser temperature increases gradually, it shows a decreasing trend in the COP of the Vapour Compression Refrigeration System. This happens because the higher condenser temperature result in increased work by the compressor to compress the refrigerant to a higher pressure. This results in lowering of the overall

efficiency. At lower condenser temperatures, rejection of heat becomes easier, which results in better refrigeration effect and higher COP values. The plot indicates maintaining efficient cooling of the condenser results in better cooling results.

Insights:

- The graph indicates an inverse relationship between COP and condenser temperature.
- As condenser temperature increases, the COP decreases gradually.
- Higher condenser temperature causes an increase in compressor discharge pressure.
- Increased discharge pressure leads to higher compressor work input.
- More compressor work reduces the efficiency of the refrigeration system.
- Lower condenser temperature helps in faster and effective heat rejection.
- Efficient heat rejection improves refrigerant condensation.
- Proper condensation increases the refrigeration effect and system performance.
- High surrounding temperature can raise condenser temperature significantly.
- Blocked airflow or dirty condenser fins may also increase condenser temperature.
- Lower condenser temperature reduces energy consumption.
- Maintaining condenser cleanliness improves thermal performance.
- Better condenser cooling helps extend compressor life.
- The graph emphasizes the importance of condenser operating conditions.
- Higher COP values represent better cooling efficiency and energy savings.

2. COP vs Evaporator Temperature



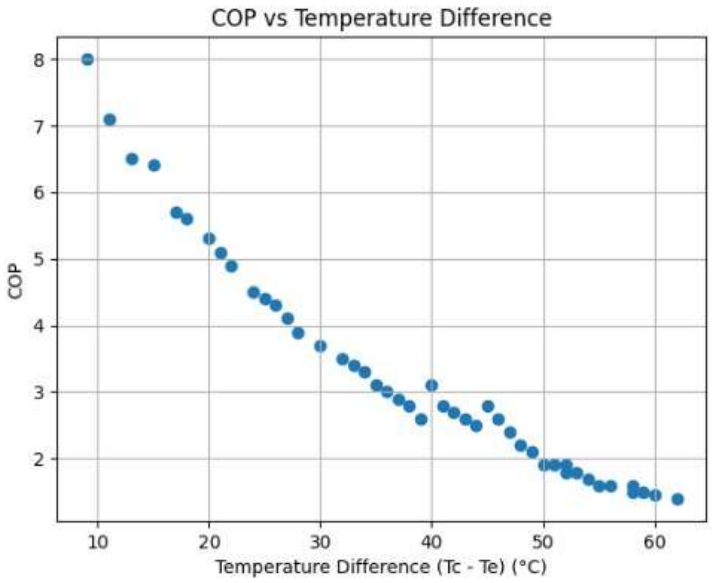
The plot between COP and Evaporator Temperature shows a direct relationship between two parameters. As there is increase in evaporator temperature, there is also increase in the COP of the system. This happens due to a higher evaporator temperature reduces the pressure difference between the evaporator and condenser, resulting in lesser work for compressor. This also results in efficiency increase of the refrigeration effect. In the converse manner, lower evaporator temperature demands higher compressor work which reduces the COP of the system. Thus, indicating the maintaining of optimum evaporator temperature is crucial for best performance in refrigeration cooling.

Insight:

- The graph shows a direct relationship between COP and evaporator temperature.
- As evaporator temperature increases, the COP also increases.
- Higher evaporator temperature reduces the pressure difference across the compressor.
- Reduced pressure difference lowers compressor work input.
- Lower compressor work improves overall system efficiency.
- Increased evaporator temperature enhances the refrigeration effect.
- Better heat absorption in the evaporator contributes to higher COP.
- Very low evaporator temperatures can significantly reduce COP.

- Lower evaporator temperature increases compressor load and power consumption.
- Operating at an optimum evaporator temperature ensures balanced performance.
- Stable evaporator temperature improves system reliability and cooling consistency.
- Proper refrigerant charge and evaporator design influence the graph trend.
- The graph highlights the importance of evaporator operating conditions in system efficiency.
- Higher COP values indicate better energy savings and effective refrigeration performance.

3. COP vs Temperature Difference

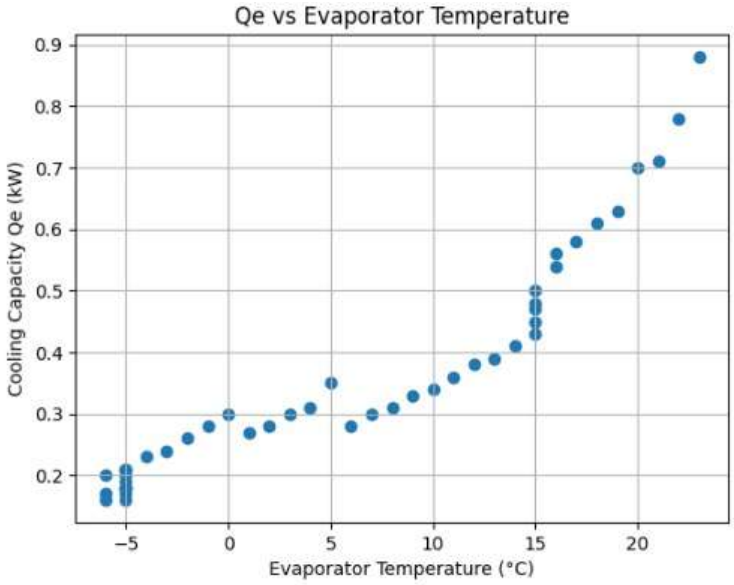


The graph between COP and Temperature Difference (difference between condenser temperature and evaporator temperature) shows an inverse relation. The COP of the Vapor Compression Refrigeration System decreases significantly with the increase of the temperature difference. This is because a larger temperature difference causes the compressor to work harder to keep the pressure ratio between the evaporator and the condenser at the required level. The increase in compressor work leads to a decrease in the overall efficiency of the system. Higher COP values are obtained when the temperature difference is smaller, and the refrigeration cycle operates more efficiently. The graph emphasizes the need for minimizing temperature difference for better performance of the system and energy saving.

Insight:

- The graph shows an inverse relationship between COP and temperature difference.
- As the temperature difference increases, the COP decreases.
- A larger temperature difference means the compressor must work harder.
- Increased compressor work leads to higher power consumption.
- Lower COP indicates reduced refrigeration system efficiency.
- Temperature difference usually refers to the gap between condenser and evaporator temperatures.
- Higher condenser temperature can increase the temperature difference.
- Lower evaporator temperature can also raise the temperature difference.
- Smaller temperature difference allows smoother refrigerant cycle operation.
- Reduced temperature difference improves refrigerating effect and system performance.
- Efficient heat rejection at the condenser helps maintain a lower temperature difference.
- Proper evaporator operation also contributes to better COP values.
- The graph highlights the importance of balanced operating temperatures.
- Maintaining minimum temperature difference improves energy savings and compressor life.

4. Q_e vs Evaporator Temperature



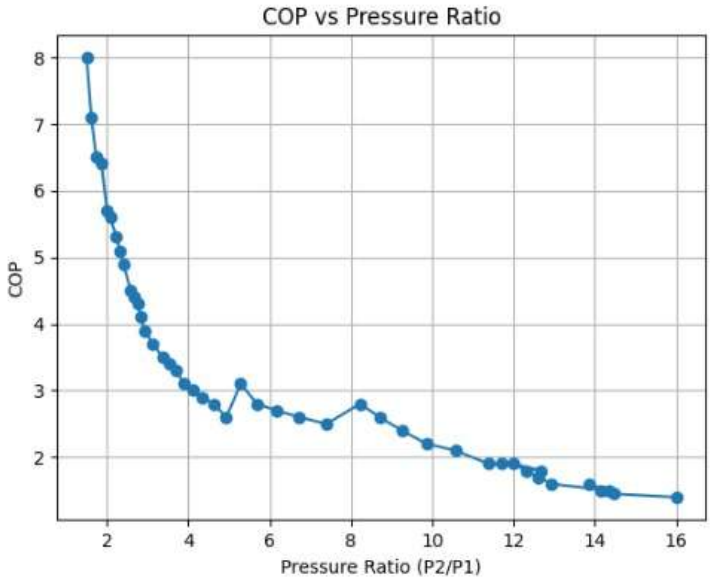
Graph between Evaporator Temperature and Cooling Capacity (Qe) shows a direct relationship between the two variables. The cooling capacity of the Vapor Compression Refrigeration System is in general increased with the increase of the evaporator temperature. This is because of the higher evaporator temperatures which allow higher refrigerant mass flow rate and better heat absorption in the evaporator which results in better refrigeration effect. At very low evaporator temperatures the compressor has to work at higher pressure differences which might lead to a reduction of the system capacity. Running at an optimum evaporator temperature helps maximize cooling output while keeping the system running efficiently and stably.

Insight:

- The graph shows a direct relationship between cooling capacity (Qe) and evaporator temperature.
- As evaporator temperature increases, the cooling capacity generally increases.
- Higher evaporator temperature improves refrigerant heat absorption in the evaporator.
- Increased evaporator pressure can enhance refrigerant mass flow rate.
- Better refrigerant flow contributes to higher cooling output.

- Lower evaporator temperatures may reduce cooling capacity due to increased compressor load.
- Extremely low evaporator temperatures can lower system efficiency.
- Optimum evaporator temperature helps maintain balanced performance and energy use.
- Higher cooling capacity leads to faster water chilling or space cooling.
- Stable evaporator temperature improves consistency of system operation.
- Proper evaporator design and refrigerant charge influence the graph trend.
- The graph highlights the importance of selecting suitable evaporator operating conditions.
- Efficient evaporator performance improves overall refrigeration system effectiveness.

5. COP vs Pressure Ratio



In a Vapor Compression Refrigeration System, the plot of COP vs Pressure Ratio is inversely proportional. Pressure ratio is defined as Compressor discharge pressure upon Compressor suction pressure. With increasing pressure ratio, the compressor has to do more work to raise the pressure of the refrigerant from the low pressure in the evaporator to the high pressure in the condenser. This additional work input reduces the overall efficiency of the refrigeration

cycle, leading to a lower COP. At lower pressure ratios the compressor is more efficient and consumes less power, and the refrigerating effect is comparatively higher, thus resulting in better COP values. The high-pressure ratio can be due to increase in condenser temperature, decrease in evaporator temperature or poor heat transfer in the condenser.

Insights:

- The graph shows an inverse relationship between COP and pressure ratio.
- As pressure ratio increases, the COP of the refrigeration system decreases.
- Higher pressure ratio means the compressor requires more work input.
- Increased compressor work reduces overall system efficiency.
- Lower pressure ratio results in better refrigerating effect and improved COP.
- High condenser pressure can increase the pressure ratio significantly.
- Low evaporator pressure also contributes to higher pressure ratio.
- Poor condenser cooling may lead to reduced COP by increasing discharge pressure.
- Maintaining optimum evaporator and condenser temperatures helps control pressure ratio.
- Lower pressure ratio operation reduces power consumption and compressor load.
- Stable pressure ratio improves reliability and lifespan of the compressor.
- The graph highlights the importance of proper heat transfer and system maintenance.
- Efficient operating conditions lead to higher COP and better energy savings.

CHAPTER 5

SUMMARY AND CONCLUSIONS

5.1 Summary of Work

The project successfully designed and fabricated an operational Vapour compression based mobile water chilling unit using a refrigerator compressor, water dispensing unit, condenser and R-600a refrigerant. The system demonstrated the principles of refrigeration cycle and practical applications of thermal engineering.

The water chilling unit was provided mobility using a cart fabricated with the added components of a battery and an inverter which can act as backup source of power also can be used in future scope of developments to this project.

5.2 Conclusions

The project successfully demonstrated the design, fabrication and operation of a VCRS system applied to a mobile water chilling unit concept. The project used a household refrigerator compressor, copper tubing network, filter dryer, condenser and R-600a as refrigerant for water chilling applications. The system operated efficiently and effectively under normal ambient conditions and insured the application and implementation of refrigeration cycle principles using low-cost, recyclable and readily available components.

One of the conclusions was the usage of spiral condenser proved to be effective in rejecting heat due to its large surface area of construction. This resulted in an improvement in condensation process. The water dispensing unit functioned as an evaporator chamber, achieving noticeable temperature reduction in water within a reasonable time period.

The hydrocarbon refrigerant R-600a (Isobutane) showed desirable cooling performance with comparatively lower energy consumption, making it as an suitable alternative while being environment-friendly. Its zero Ozone Depletion Potential while minimal Global Warming Potential enhance the sustainability of the designed system.

The project additionally established the recycled or re-usable components can be inculcated to design and fabricate a system which can provide the necessary cooling as per requirement

within a reduced budget. This makes the system economic by cost while also viable for educational-experimental purposes and cooling applications in small households and small industries.

As a whole, the apparatus offers an efficient, eco-friendly, and practical solution for water chilling while promoting design innovation, resource optimization and sustainable refrigeration practices.

5.3 Future Scope of Work

The developed Vapour compression based mobile water chilling unit offers significant opportunities for further improvement and advanced research:

1. The enhancement of condenser performance by installing a forced air-cooling fan over the condenser. This can help in improving the heat rejection, reduce compressor workload and increase the overall coefficient of performance of the system.

2. Integration of a digital thermostat and automatic temperature control system which can maintain precise water temperature and reduce unnecessary compressor cycling. This would result in lesser energy wastage and better user convenience.

3. Upgradation of the system with IoT-based sensors and smart monitoring, enabling real time observation of temperature, pressure, consumption of power and fault diagnosis through mobile applications or any web platform.

4. The use of renewable sources of energy like solar energy by integration of solar photovoltaic panels to power the compressor and charge the battery can be another major future scope of work. This can further expand its applications in rural areas where electrical energy is in deficit.

5. Performance analysis with other hydrocarbon refrigerants which may result in better performance alternative than the current R-600a refrigerant.

6. Autonomous mobility integration by inculcating ultrasonic sensors and obstacle detection and obstacle avoiding code to give free movement to the water chilling without any human intervention.

APPENDIX II

BILL OF MATERIALS

S.no.	Items	Cost
1	Mild Steel Plate for Mounting	400
2	Copper tube, Gas Accessories	1600
3	Compressor	1000
4	Condenser and Mounting Bolts	500
5	Water Drum + Tap	400
6	Cart Fabrication	4000
7	Wheels	400
Total Cost		8300

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